# KARTHIK CHAKRAVARTULA

**Senior Software Engineer | Solutions Architect | AI, Dynamics 365, .NET and Azure**

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## PROFESSIONAL SUMMARY

Senior software engineer and solutions architect with 14+ years of building and deploying software inside customer environments. Spent eight years as a consultant working directly with clients including Wells Fargo, Pioneer (DuPont) and Alameda County, turning their business problems into production systems. Now designs agentic AI and generative AI (GenAI) solutions with RAG, Model Context Protocol (MCP), Python and LLM evals, alongside Dynamics 365, .NET and Azure platforms at Farm Credit Bank of Texas. Suited to forward deployed engineering: works directly with stakeholders, prototypes quickly, integrates with legacy and third-party systems, and stays through deployment.

## CORE SKILLS

**AI and LLMs:** Agentic AI, generative AI (GenAI), large language models (LLMs), retrieval-augmented generation (RAG), Model Context Protocol (MCP), AI orchestration, tool calling, prompt engineering, LLM evaluation (evals), Microsoft Copilot

**Languages:** Python, TypeScript, JavaScript, C#, SQL (T-SQL), PowerShell

**Application development:** .NET, .NET Core, ASP.NET MVC, Web API, REST APIs, React, Angular, microservices, full-stack development

**Cloud and integration:** Microsoft Azure, Azure Functions, Azure API Management, Azure Blob Storage, Microsoft Entra ID, OData, third-party and legacy system integration

**Enterprise platforms:** Microsoft Dynamics 365 CE, Dynamics CRM 2011/2013/2016, Power Platform, Dataverse, Power Automate, Power Pages, PCF controls

**Data:** SQL Server, MongoDB, Oracle, SSIS, SSRS, Power BI, Entity Framework

**Delivery:** Azure DevOps, CI/CD, Git, GitHub, Octopus Deploy, Jenkins, Agile, test-driven development (TDD)

**Customer-facing:** Client consulting, requirements discovery, stakeholder communication, solution design, rapid prototyping, production deployment, mentoring

## PROFESSIONAL EXPERIENCE

**Farm Credit Bank of Texas, Austin, TX** **07/2021 – Present**

*Senior Software Engineer*

- Design and prototype agentic AI and GenAI solutions using RAG and Model Context Protocol (MCP) pipelines that connect LLMs to enterprise data and tools, with Python services and LLM evals to measure quality before release.
- Architected the integration of Dynamics 365 CE with the bank's loan origination system, extending the platform with plugins, workflows and Power Platform solutions.
- Deploy and configure solutions for multiple lending associations, each with its own Microsoft Entra tenant, environment and release pipeline.
- Built borrower portal sign-in on Microsoft Entra External ID, with custom authentication extensions in C# Azure Functions behind Azure API Management.
- Designed audit-log archival to Azure Blob Storage that saved thousands of dollars in cloud storage costs, with a React app embedded in the CRM so users can still look up history.
- Automated deployment of Dataverse solutions and Power Pages sites with Azure DevOps YAML pipelines and PowerShell.
- Mentor engineers, review code and delegate work across the team to meet project deadlines.

**Collaborate Solutions** **01/2013 – 07/2021**

*Software Engineer*, *Consultant*

- Served as a key liaison with client teams on planning, strategy and design for product releases across banking, agriculture and public-sector engagements.

***Farm Credit Bank of Texas*** *2018 – 2021*

- Built .NET APIs that integrate third-party tools with Dynamics CRM, including an API that generates customer numbers from Encompass and creates the matching account in CRM.

- Built a self-service application that lets users activate and deactivate their own accounts.

***Wells Fargo*, *Full Stack .NET Developer*** *2016 – 2018*

- Replaced the legacy SMC customer-data application and built a tool that connects to third-party services for instant data access.
- Automated data extraction from IBM 3270 mainframe screens using .NET, and built an Angular application for grid-based data editing.

***Pioneer (DuPont)*, *Software Developer*** *2015 – 2016*

- Built the enhancement engine for the Encirca platform, exposing farmer crop-yield services through ASP.NET Web API across multiple databases.
- Set up Octopus Deploy for automated deployments and tuned stored procedures and triggers for performance.

***Alameda County*, *Oakland*, *CA*, *Systems Engineer*** *2014*

- Gathered business rules with the HR department and rebuilt a legacy COBOL position-requisition system as a single-page web application.
- Built a mobile web app with ESRI maps to locate parks, parking lots and libraries from public datasets.

***Wells Fargo*, *Applications Systems Engineer*** *2013 – 2015*

- Migrated ASP.NET Web Forms applications to ASP.NET MVC with REST services, implemented security certificates for third-party web services, and rebuilt reporting in SSRS.

**xRM.com** **05/2012 – 12/2012**

*CRM Developer*

- Extended Microsoft Dynamics CRM 2011 to manage employee payroll, and built an adapter that automatically creates support cases from incoming faxes.

## PROJECTS AND COMMUNITY

- D365 Field Editor and D365 JS Library Editor: free Chrome extensions for Dynamics 365 developers, published on the Chrome Web Store.
- Published a PCF control on PCF Gallery; articles on DZone and MSDynamicsWorld.
- Volunteer engineer for Schoolgroup.org, U.S. Digital Response, NYC DSSG and Vitas (.NET, Dynamics 365, Salesforce, HubSpot, React, TypeScript, Python).

## EDUCATION

**Master's in Electrical and Computer Engineering**, Southern Illinois University Carbondale, IL

**Bachelor's in Electronics and Communications Engineering**, Jawaharlal Nehru Technological University, Hyderabad, India

## CERTIFICATIONS

- Microsoft Power Platform Developer certification
- Microsoft Certified Technology Specialist (MCTS)
- Microsoft Dynamics CRM Customization and Configuration; Extending Microsoft Dynamics CRM